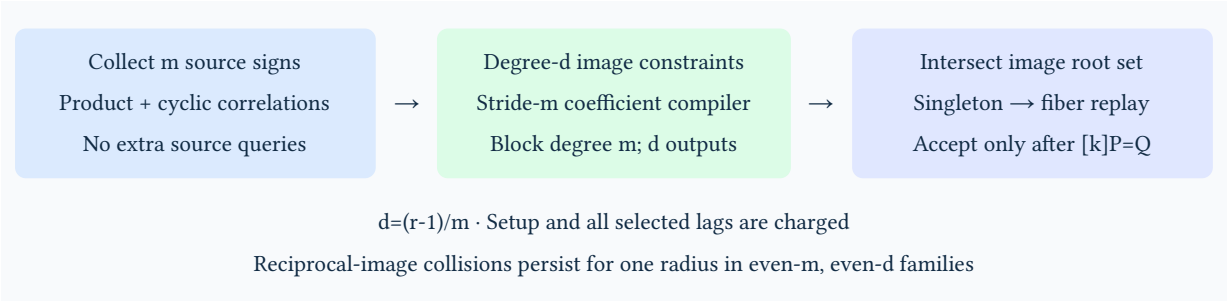
Cyclic correlations on power images

T-1808 • 2026-11-07 • PCC-20261107-c6c777 • unsigned producer evidence

The image phase now uses cyclic correlations of individual orbit signs, in addition to their product. These provide polynomial constraints on z=(k+c)^m before the image becomes a singleton. They require no extra source queries once the orbit has been collected.



Feature and coefficient compiler

Let s_j=χ(k+c−bζ^j), a=ζ^ell, h=(r−1)/2 and d=(r−1)/m. If ((X−1)(X−a))^h has coefficients c_n, then the cyclic correlation sum of s_j s*(j+ell) equals C(z/b^m), where C(V)=m sum_t c*(tm)V^t. The root-of-unity sum cancels all other exponents; the factored source multiplier is a square. The identity includes zero sign entries.

The coefficient recurrence is a(j+1)c*(j+1)=(j−h)(1+a)c_j+(2h−j+1)c*(j−1). A balanced2×2 polynomial-matrix block advances m indices. Evaluating it at0,m,…,(d−1)m retrieves only the needed coefficients. All denominators lie between indices1 and r−1 and remain invertible. A single explicit batch inverse normalizes the evaluated blocks. Production does not expand the degree-r−1 coefficient polynomial.

Exact native checks

Final validation-02 passes four selected methods,157 imported filtered.98 compilers match independently expanded schoolbook powers across primes7,13,17,31,41,97, divisors m from2 through8 and all lags.12,048 direct orbit values agree.2,691 complete image sets match the intersections;1,434 are strictly refined by correlations.618 quadratic cases retain exactly the product-only image set.

Fifteen larger cases at r65537 use block widths64,128,256,512,1024 and lags0,1,m/2. They match a streamed scalar recurrence, exercising Karatsuba and Newton arithmetic. This larger reference shares the recurrence identity; the small dense references do not.

Public recovery with lag1 passes all46 fixtures: frozen20 on E/F509, $y^2=x^3+2x+10$, P=(3,68), r479,m2; all13 on E/F7, $y^2=x^3+3$, P=(1,2), r13 with m3 and m6.28 use retained-sign replay.453 source calls are recorded as software coverage, not a speed result.32 source/input hashes and generation hashes match. Original4,563 cohort unchanged; no cloud or timing panel.

Symmetry that the extractor must retain

The coefficients obey C[t]=χ(a)C[d−t]. For nonzero normalized images v with v^d=1, C(1/v)=χ(a)C(v). When m and d are both even, all correlations and the folded product agree for reciprocal images at a fixed radius.792 ordered controls include660 cases with distinct reciprocal images. These are retained collisions, not silently accepted recoveries. Another radius changes the reciprocal pairing; no universal separation bound is supplied. Pair correlation at m2 is redundant.

Cost and accounting correction

One lag has a safe setup bound O(M(m+d) log(m+d)+log r), using installed Karatsuba multiplication and ordinary product/remainder trees. Balanced m,d of order sqrt(r) give the derived exponent log_2(3)/2, with a suitable divisor required. L lags multiply setup and stored coefficients; compiling all m lags is not free. Conservative compiler workspace is O(m log(m+1)+d log(d+1)). Each orbit still uses m separate source evaluations and their group shifts.

General polynomial-backend inverses use extended Euclid. Earlier reports incorrectly described them as exponentiation-based; historical evidence is preserved and INVERSION_CORRECTION.md fixes the accounting. This compiler's explicit batch inverse and scalar powers do use exponentiation, while monic tree reciprocals skip unit inversions. Later gcd inversions remain separately charged.

Proof map, sources and next construction

Lean4.34.1 checks the sequential block-composition invariant with no axioms. The finite-field projection, recurrence interpretation, symmetry and complexity remain written proofs. Validation-01 source snapshots are retained before symmetry controls. The report is producer evidence without an independent signed verdict.

Sources: existing blocked recurrence, inert seed and fold/fiber modules; Bostan–Gaudry–Schost2007, DOI10.1137/S0097539704443793, introduction/main algorithm and coefficient application. Established recurrence acceleration is reused; no novelty claim. KB unavailable. Next: exploit reciprocal structure or higher orbit invariants without compiling every lag, while resolving image separation across radii and retaining source cost.